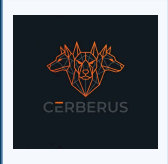


Treasury Auction Demand Stress



INTERNAL

Treasury auction demand stress study measuring tail, bid-to-cover, and forward SPY/TLT returns.

Author: Cerberus Research

Generated: 2026-08-07 08:47 UTC

N 1,326	EDGE +1.27%	HIT-RATE 69.5%
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CLASSIFICATION internal	AUTHOR Cerberus Research
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RUN ID 20260807_094716	GIT COMMIT 978cf61e
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DURATION 2.7s	DATASETS treasury/treasury_auction_features
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PARAMETER · PERIOD 10y	PARAMETER · HORIZON 21
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KPI Value	N 1,326
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EDGE +1.27%	HIT-RATE 69.5%
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KEY FINDINGS · Latest demand score: 0.000
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Data Sources

- **treasury/treasury_auction_features**: Treasury auction features including tail, BTC, and size z-scores

Key Findings

- Latest demand score: 0.000

Executive Summary

NOTE

Treasury auction demand stress study. Combines auction tail and bid-to-cover z-scores into a composite demand score, then tests forward SPY and TLT returns across strong and weak demand regimes.

Results & Interpretation

- Across 1,326 events, the forward return averaged +1.27%. Returns were positive 69.5% of the time.
- Across 452 events, the forward return averaged +1.11%. Returns were positive 67.0% of the time.
- Across 1,328 events, the forward return averaged -0.45%. Returns were positive 43.1% of the time.
- Across 455 events, the forward return averaged -0.37%. Returns were positive 47.0% of the time.

Objectives

NOTE

Evaluate whether the studied signal produces a measurable, repeatable edge.

Quantify the reliability of the observed patterns across the sample.

Methodology

NOTE

Daily metrics are averaged across all auctions. Demand score = -tail z-score + BTC z-score. Regimes are split at the median demand score. Forward returns are computed over the configured horizon.

Charts

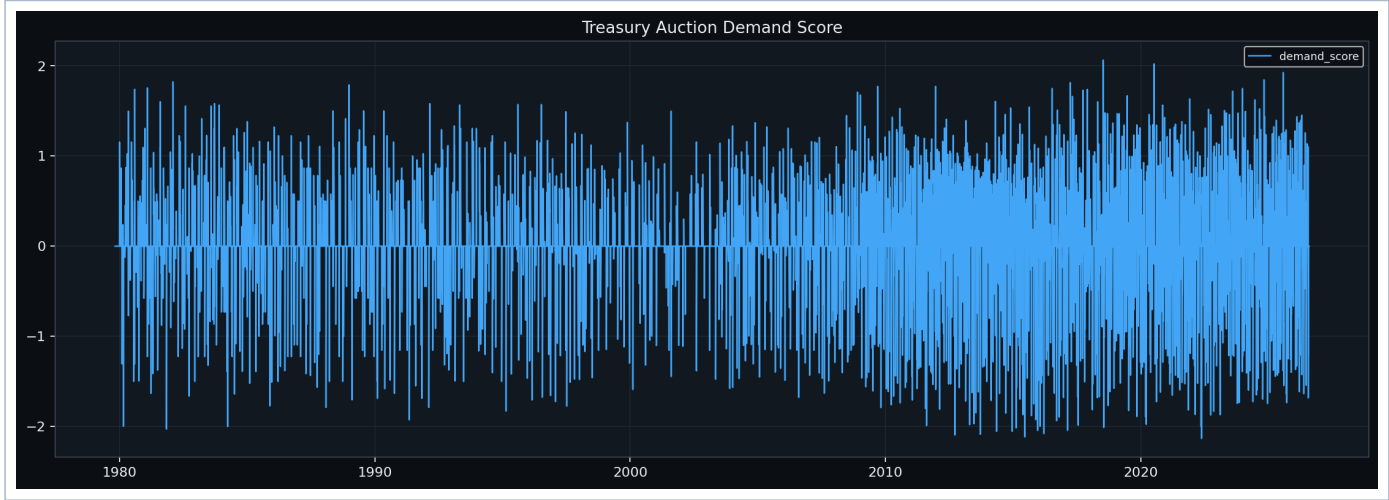


Figure 1. Auction demand score
Auction demand score

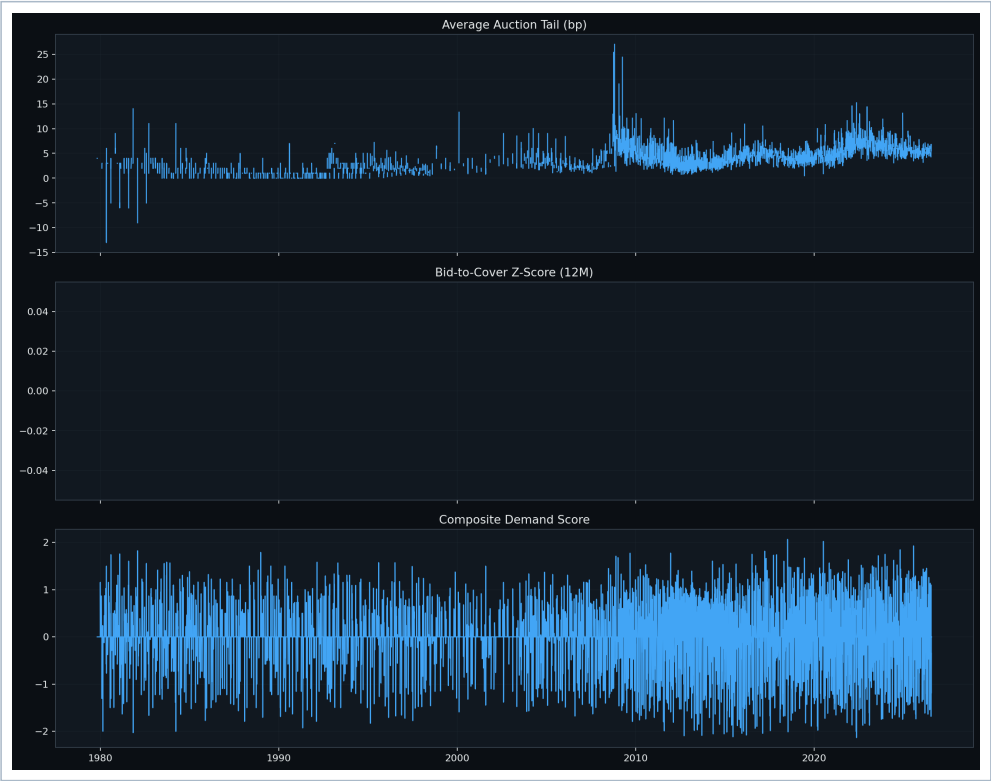


Figure 2. Auction demand dashboard
Auction demand dashboard

Statistical Results

tail_bp	tail_score	demand_sc...	weak_dema...	strong_demand
6.00	-0.93	0.93	False	True
8.70	0.40	-0.40	True	False
5.50	-1.40	1.40	False	True
6.70	-0.59	0.59	False	True
5.80	-1.02	1.02	False	True

tail_bp	tail_score	demand_sc...	weak_dema...	strong_demand
5.50	-0.71	0.71	False	True
6.10	-0.20	0.20	False	True
7.00	1.43	-1.43	True	False
7.30	1.00	-1.00	True	False
8.50	1.40	-1.40	True	False
5.50	-1.20	1.20	False	True
6.10	-0.68	0.68	False	True
5.00	-1.22	1.22	False	True
5.30	-0.72	0.72	False	True
4.30	-1.17	1.17	False	True
4.40	-0.83	0.83	False	True
6.20	1.26	-1.26	True	False
4.30	-0.96	0.96	False	True
5.60	0.73	-0.73	True	False
6.00	1.02	-1.02	True	False
6.20	1.08	-1.08	True	False
5.50	-0.03	0.03	False	True
5.20	-1.25	1.25	False	True
5.10	-1.15	1.15	False	True
4.90	-1.10	1.10	False	True
5.30	0.45	-0.45	True	False
5.65	1.38	-1.38	True	False
3.50	-1.93	1.93	False	True
5.90	0.90	-0.90	True	False
6.00	0.85	-0.85	True	False
7.30	1.55	-1.55	True	False
5.60	-0.80	0.80	False	True
5.00	-1.14	1.14	False	True
4.10	-1.26	1.26	False	True
5.70	0.14	-0.14	True	False
5.50	0.48	-0.48	True	False
4.00	-1.29	1.29	False	True

Research only. Not investment advice. Past performance is not indicative of future results.

tail_bp	tail_score	demand_sc...	weak_dema...	strong_demand
4.70	-0.34	0.34	False	True
6.60	1.63	-1.63	True	False
4.90	-0.18	0.18	False	True
8.60	1.74	-1.74	True	False
5.10	-0.33	0.33	False	True
5.60	-0.03	0.03	False	True
5.80	0.10	-0.10	True	False
4.20	-1.01	1.01	False	True
5.70	-0.09	0.09	False	True
6.40	1.25	-1.25	True	False
4.20	-0.84	0.84	False	True
6.00	0.68	-0.68	True	False
5.05	-0.22	0.22	False	True
5.50	0.24	-0.24	True	False
3.90	-1.17	1.17	False	True
4.60	-0.35	0.35	False	True
6.40	1.26	-1.26	True	False
5.60	0.43	-0.43	True	False
7.80	1.40	-1.40	True	False
5.50	-0.10	0.10	False	True
4.70	-0.62	0.62	False	True
6.60	0.76	-0.76	True	False
4.40	-1.08	1.08	False	True
5.50	-0.20	0.20	False	True
5.30	-0.33	0.33	False	True
5.20	-0.25	0.25	False	True
5.40	0.55	-0.55	True	False
5.40	0.49	-0.49	True	False
5.50	0.66	-0.66	True	False
4.90	-0.79	0.79	False	True
4.85	-1.16	1.16	False	True
4.50	-1.28	1.28	False	True

Research only. Not investment advice. Past performance is not indicative of future results.

tail_bp	tail_score	demand_sc...	weak_dema...	strong_demand
4.00	-1.05	1.05	False	True
6.00	1.37	-1.37	True	False
4.00	-0.81	0.81	False	True
5.30	0.67	-0.67	True	False
5.80	1.08	-1.08	True	False
5.00	-0.02	0.02	False	True
6.70	1.33	-1.33	True	False
3.20	-1.44	1.44	False	True
6.90	0.92	-0.92	True	False
4.30	-0.71	0.71	False	True
5.20	-0.01	0.01	False	True
6.00	0.51	-0.51	True	False
5.00	-0.23	0.23	False	True
5.20	-0.26	0.26	False	True
5.70	0.79	-0.79	True	False
4.60	-1.37	1.37	False	True
4.70	-0.90	0.90	False	True
7.10	1.61	-1.61	True	False
7.10	1.21	-1.21	True	False
5.70	-0.03	0.03	False	True
6.50	0.68	-0.68	True	False
5.00	-1.39	1.39	False	True
5.20	-0.97	0.97	False	True
5.30	-0.60	0.60	False	True
4.20	-1.45	1.45	False	True
6.20	1.43	-1.43	True	False
5.75	0.70	-0.70	True	False
5.50	0.21	-0.21	True	False
4.80	-0.62	0.62	False	True
5.50	0.25	-0.25	True	False
5.60	0.09	-0.09	True	False
5.40	0.21	-0.21	True	False

tail_bp	tail_score	demand_sc...	weak_dema...	strong_demand
6.90	1.64	-1.64	True	False
5.60	-0.05	0.05	False	True
5.70	0.09	-0.09	True	False
5.00	-0.90	0.90	False	True
4.50	-1.26	1.26	False	True
5.80	0.33	-0.33	True	False
6.00	0.47	-0.47	True	False
5.20	-0.29	0.29	False	True
5.00	-0.49	0.49	False	True
5.10	-0.30	0.30	False	True
6.50	1.54	-1.54	True	False
6.00	0.73	-0.73	True	False
4.50	-1.14	1.14	False	True
5.00	-0.47	0.47	False	True
5.10	-0.36	0.36	False	True
4.30	-1.10	1.10	False	True
6.80	1.68	-1.68	True	False
5.50	0.34	-0.34	True	False
6.80	1.19	-1.19	True	False

Table 1. Daily auction metrics

regime	count	avg_fwd_ret_...	median_fwd_ret_...	hit_rate_pct
weak_demand	1326	+ 1.27%	+ 1.77%	+ 69.46%
strong_demand	452	+ 1.11%	+ 1.64%	+ 67.04%

Table 2. SPY forward returns by demand regime

regime	count	avg_fwd_ret_...	median_fwd_ret_...	hit_rate_pct
weak_demand	1328	-0.45%	-0.62%	+ 43.07%
strong_demand	455	-0.37%	-0.30%	+ 47.03%

Table 3. TLT forward returns by demand regime

Limitations

NOTE

Results are research-grade, not investment advice. The sample is limited to available data, and measured edges may not persist out-of-sample.

Appendix Tables

daily

date	tail_bp	tail_score	demand_sc...	weak_dema...	strong_demand
2025-04-16	6.00	-0.93	0.93	False	True
2025-04-17	8.70	0.40	-0.40	True	False
2025-04-22	5.50	-1.40	1.40	False	True
2025-04-23	6.70	-0.59	0.59	False	True
2025-04-24	5.80	-1.02	1.02	False	True
2025-05-05	5.50	-0.71	0.71	False	True
2025-05-06	6.10	-0.20	0.20	False	True
2025-05-08	7.00	1.43	-1.43	True	False
2025-05-21	7.30	1.00	-1.00	True	False
2025-05-22	8.50	1.40	-1.40	True	False
2025-05-27	5.50	-1.20	1.20	False	True
2025-05-28	6.10	-0.68	0.68	False	True
2025-05-29	5.00	-1.22	1.22	False	True
2025-06-10	5.30	-0.72	0.72	False	True
2025-06-11	4.30	-1.17	1.17	False	True

spy_fwd_returns

regime	count	avg_fwd_ret_...	median_fwd_ret_...	hit_rate_pct
weak_demand	1326	+ 1.27%	+ 1.77%	+ 69.46%
strong_demand	452	+ 1.11%	+ 1.64%	+ 67.04%

tlt_fwd_returns

regime	count	avg_fwd_ret_...	median_fwd_ret_...	hit_rate_pct
weak_demand	1328	-0.45%	-0.62%	+ 43.07%
strong_demand	455	-0.37%	-0.30%	+ 47.03%

Statistics Summary

Key	Value
study	treasury_auction_demand_stress
category	macro
status	success

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Key	Value
parameters	{'period': '10y', 'horizon': 21}
datasets	['treasury/treasury_auction_features']
rows	11063
git_commit	978cf61eefd9b125a5087073e70ef1c9d91f687b
executed_at	2026-08-07T08:47:16.558863+00:00

Glossary

Term	Definition
n / count	Number of observations or events in the sample.
edge	Average excess return versus the unconditional baseline.
hit-rate	Share of events with a positive forward return.
p-value	Probability of observing the result under the null of no edge.
*** / ** / *	Significance at 99.9% / 99% / 95% confidence.
α (alpha)	Intercept or active return versus benchmark.
β (beta)	Sensitivity to a reference factor or market.
σ (sigma)	Volatility / standard deviation of returns.

Document Control

Field	Value
Report schema	2
Framework version	0.1.0
Classification	internal
Generated	2026-08-07 08:47 UTC
Author	Cerberus Research
Study slug	treasury_auction_demand_stress
Run ID	20260807_094716
Git commit	978cf61e
Datasets	treasury/treasury_auction_features
Parameters hash	453adde0cf1b
Parameter · period	10y
Parameter · horizon	21
Data fingerprint	3da3cbeb3a5b
Duration	2.7s

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- Table 3: TLT forward returns by demand regime